A small garden. A complete experience.

One alpha, three graphics pillars. Two technical challenges worth examining.

Build a bioluminescent garden from seeds and simple rules.

PROCEDURAL GENERATION

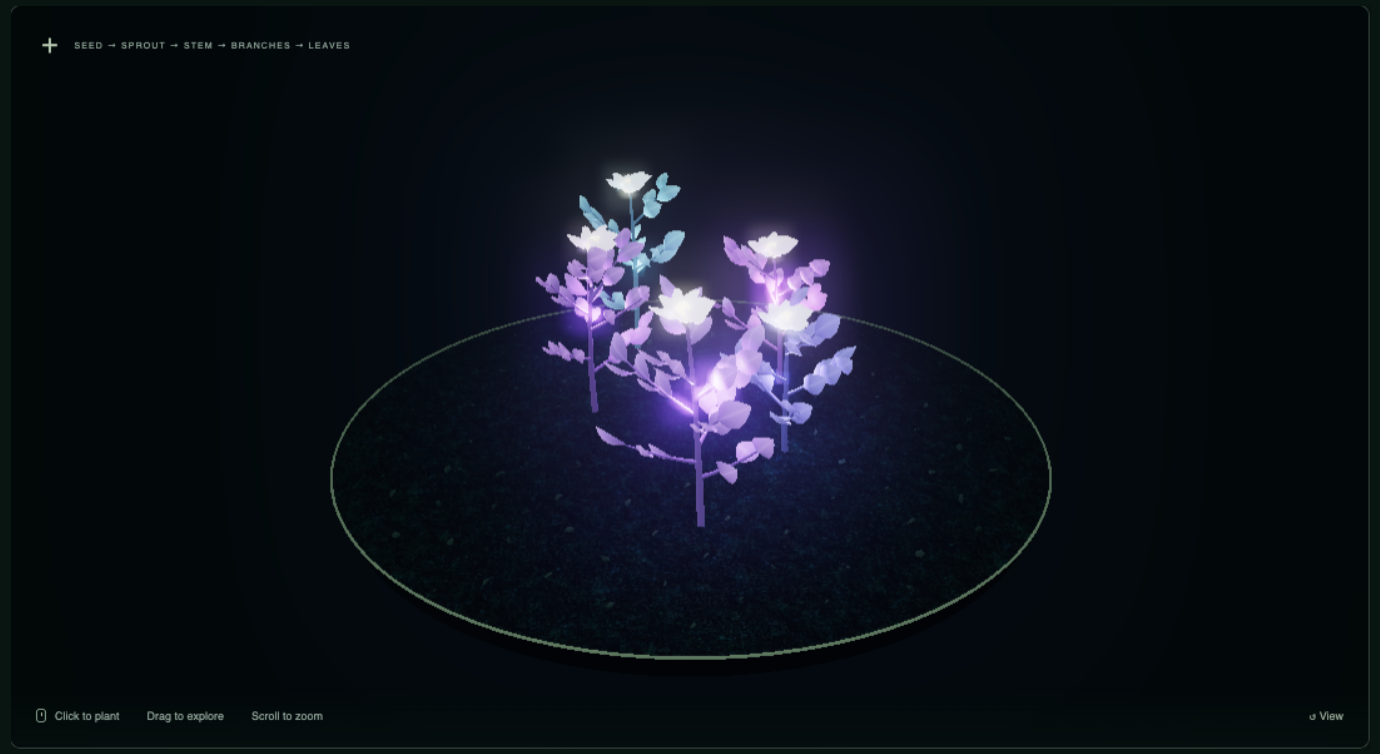
Seeded geometry and bounded variation

ANIMATION

Staged growth + coherent wind

RENDERING

Emission, colored lights, bloom + day/night



01 / START

Menu + short control guide

02 / GROW & SHAPE

Plant 1-5 seeds · wind · time of day

03 / FINISH & RESTART

Mature garden summary · clean reset

Challenge 1 / Grow structure in sequence.

A repeatable blueprint controls geometry; local timing makes each part emerge from its base.

WHY IT IS HARD

Random branches can look incoherent.
Whole-model scaling reveals every part at once instead of showing growth.

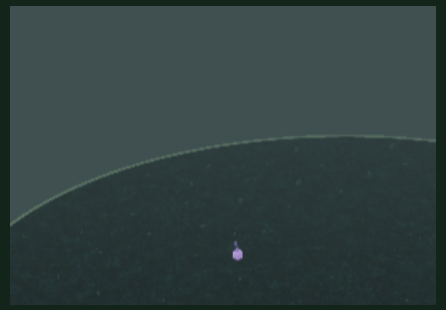
OUR APPROACH

Seed → constrained blueprint → instances.
14 trunk segments; 5-8 branches.
3-5 leaves per branch; staggered reveal.

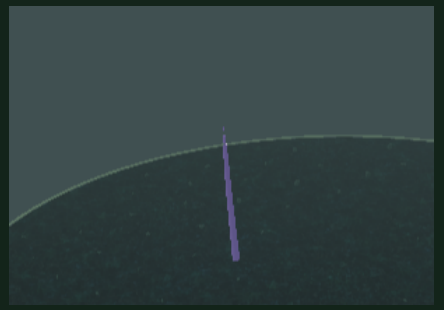
EVIDENCE + TRADEOFF

Same seed → same geometry.
300 seeds checked for finite bounds.
Intersections are not collision-resolved.

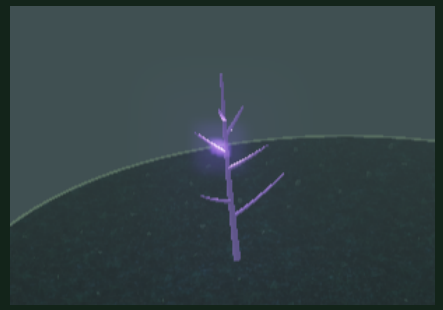
Local reveal: g = smoothstep(0, 1, clamp((progress - start) / (end - start)))



SPROUT



STEM



BRANCHES



LEAVES + BLOOM

Challenge 1 / Keep wind and growth connected.

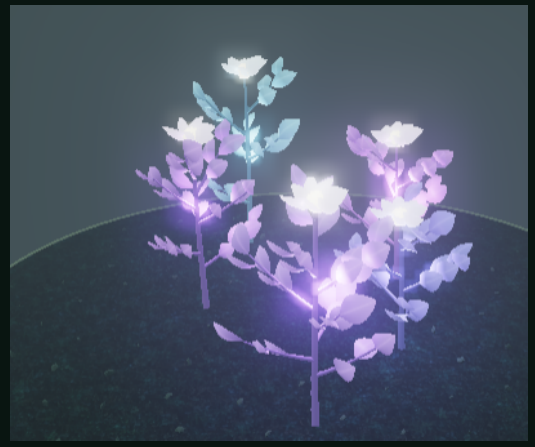
One deformation field keeps attachment positions coherent, including during staged growth.

Instance placement first.
Shared plant-space bend second.

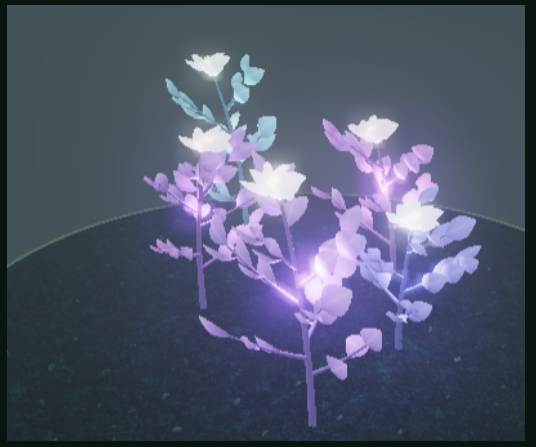
$$x' = x + kx(t) \cdot \max(y, 0)^2$$
$$z' = z + kz(t) \cdot \max(y, 0)^2$$

Root stays anchored; tips move more.
Seed-derived phase varies the sway.

Normals: inverse-transpose Jacobian.
Light anchors: same analytic CPU field.
Mature organ matrices stay unchanged.



WIND 0% / SAME LAYOUT



WIND 100% / SAME LAYOUT

COHERENCE AT LOW CPU COST

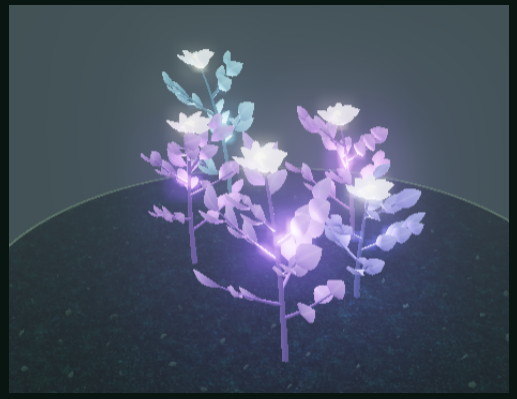
Wind updates shader uniforms, not every organ matrix.
The same field deforms attached parts continuously.
Tradeoff: no inertia, collision response, or exact length conservation.

Challenge 2 / Glow within a real-time budget.

Emission, illumination, and bloom have different jobs; profiling has to include the complete frame.



BLOOM OFF / EMISSION REMAINS



BLOOM ON / HALO RETURNS

Plants	Avg FPS	p95 ms	Calls
1	59.5	16.8	23
3	60.0	16.7	33
5	60.0	16.7	43

EMISSION

Bright plant surfaces

POINT LIGHTS

Colored light on soil

BLOOM

Screen-space halo

Daylight attenuates emission and bloom to avoid washout.
Instancing limits geometry draw calls.
FPS capped near 60 does not reveal spare headroom.

M2 / Chrome 156 / 1086 × 594 / DPR 1
Wind 70%; daylight 15%; bloom control 0.65.
10s warm-up + 8s sampling at each level.
Slowest recorded frame: 83.4 ms (included).

One GPU; no GPU timer queries.
Five plants do not validate the original 20-plant target.